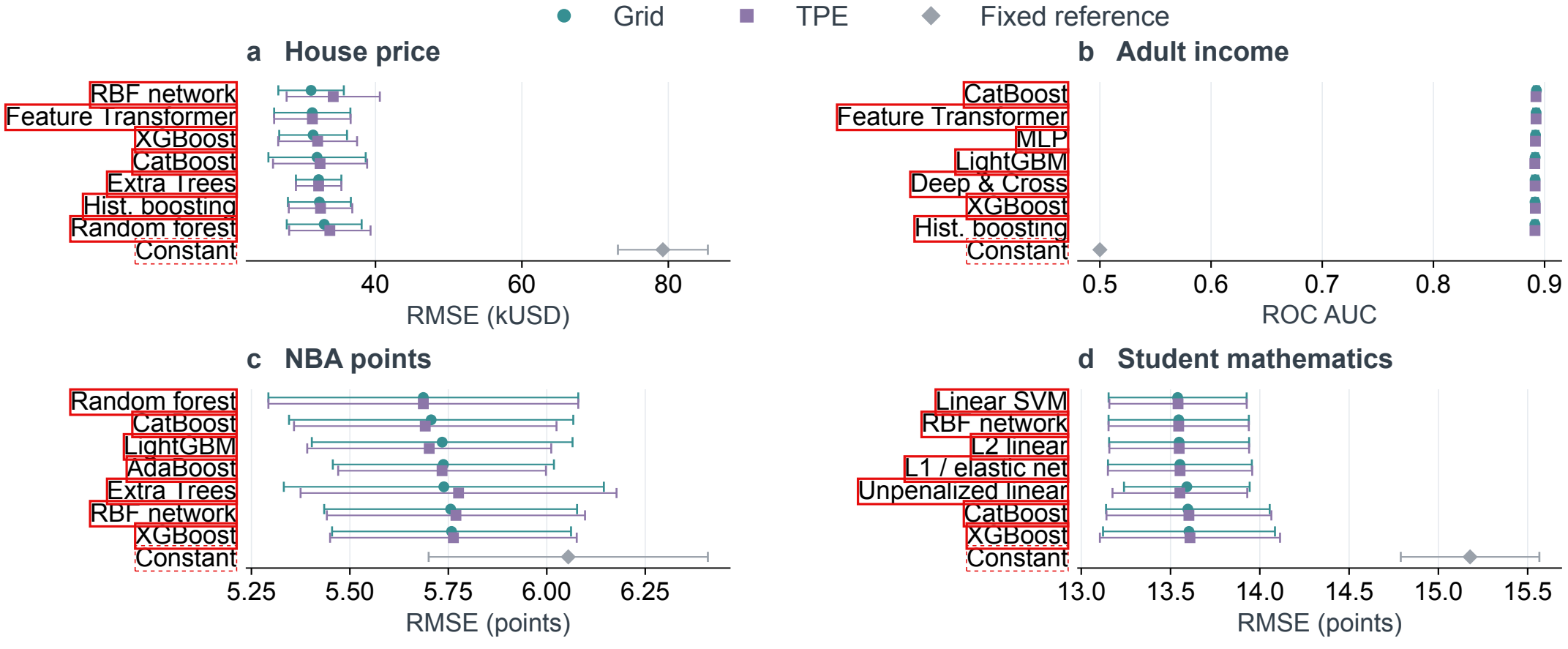

Grid
TPE
Fixed reference
a House price
RBF network
Feature Transformer
XGBoost
CatBoost
Extra Trees
Hist. boosting
Random forest
Constant
40
60
80
RMSE (kUSD)
b Adult income
CatBoost
Feature Transformer
MLP
LightGBM
Deep & Cross
XGBoost
Hist. boosting
Constant
0.5
0.6
0.7
0.8
0.9
ROC AUC
c NBA points
Random forest
CatBoost
LightGBM
AdaBoost
Extra Trees
RBF network
XGBoost
Constant
5.25
5.50
5.75
6.00
6.25
RMSE (points)
d Student mathematics
Linear SVM
RBF network
L2 linear
L1 / elastic net
Unpenalized linear
CatBoost
XGBoost
Constant
13.0
13.5
14.0
14.5
15.0
15.5
RMSE (points)